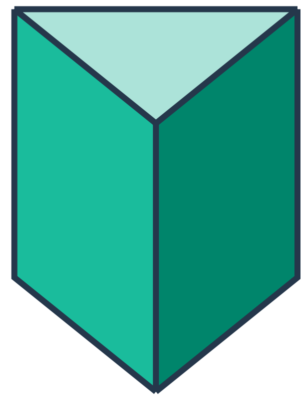
Cross sections of solids

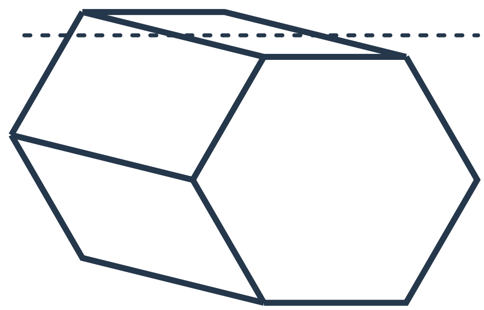
1 Consider the given solid:



a Does the shape have a uniform cross section?

b Classify the solid.

2 Consider the given solid:



a If the solid is cut straight down below the dotted line, classify the resulting cross section.

b Does the solid above have a uniform cross section?

c Classify the solid.

3 Which one of the following shapes cannot be a cross section of a triangular prism?

A Rectangle **B** Square **C** Circle **D** Triangle

4 Which of the following shapes could be a cross section of a triangular pyramid?

A Rectangle **B** Square **C** Circle **D** Triangle

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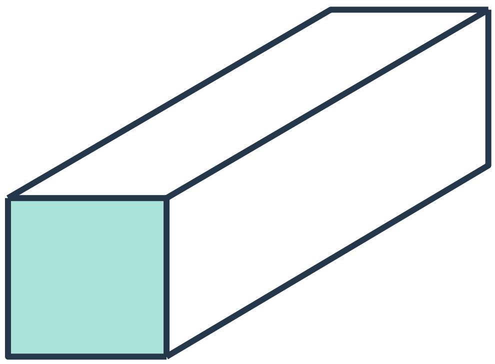
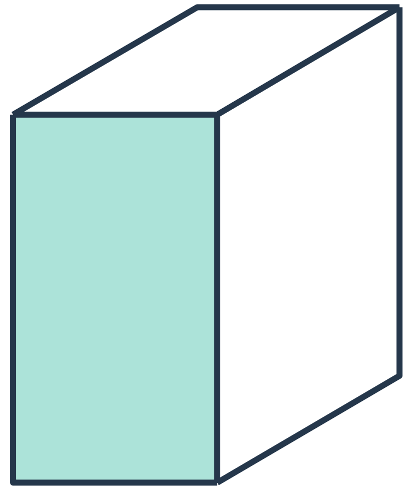
5 Suppose you are cutting each of the followi[redacted]lids parallel to their bases. Which solid does NOT have cross sections that are identical in size and shape to their shaded base when cut this way?

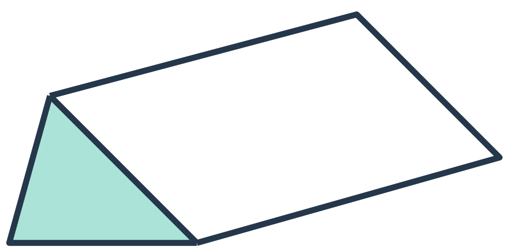
A



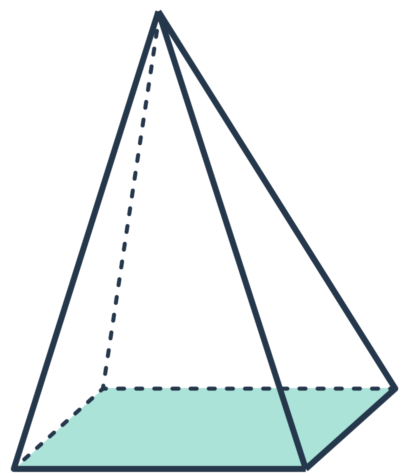
B



C



D



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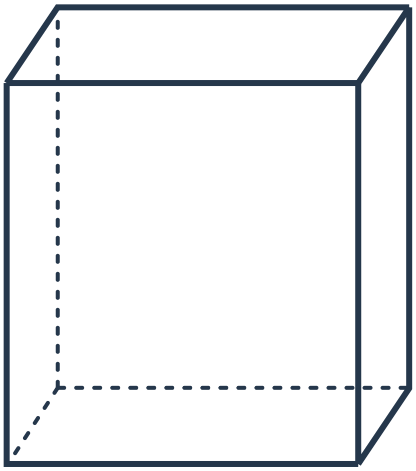
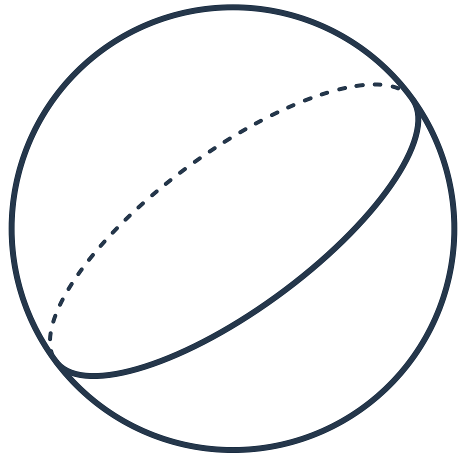
6 Identify the shape formed by the intersectio[redacted]a horizontal plane with the given solid.

a

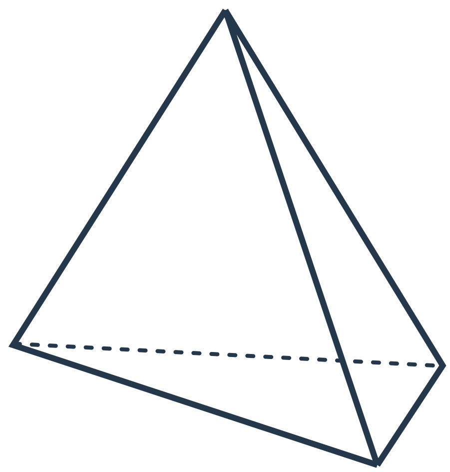


b



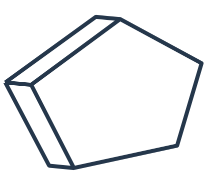
c

d

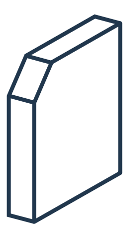


7 Sketch the cross section of the following prisms:

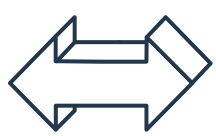
a



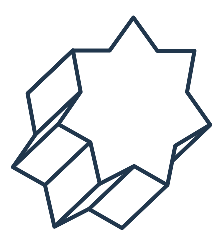
b



c



d



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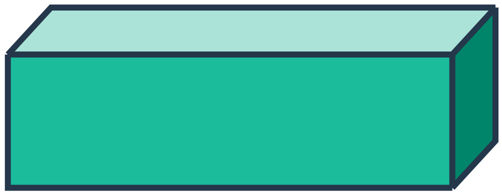
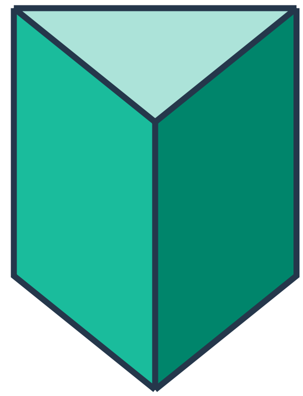
8 Sketch the cross section formed by the intersection of the described plane with the following solids:

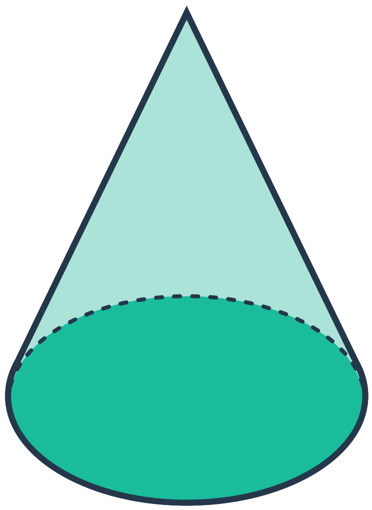
a A plane perpendicular to the prism



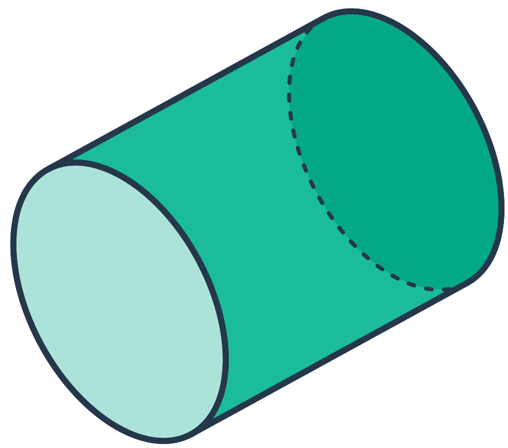
b A plane parallel to the base



c A plane parallel to the base



d A plane perpendicular to the base



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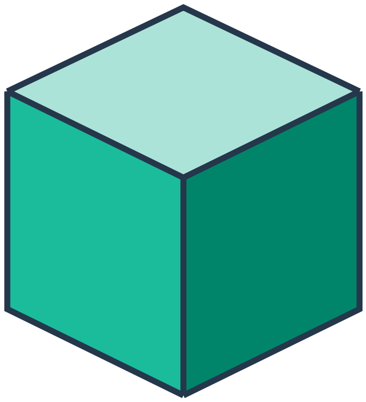
9 For each solid:

i Sketch the cross section parallel to the base

ii Sketch the cross section perpendicular to the base

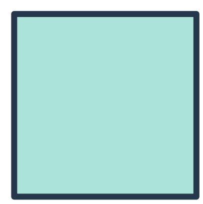
a



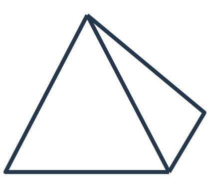
b

Solids from cross sections

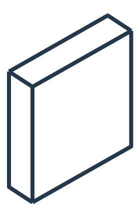
10 A solid is cut parallel to its base and the resulting uniform cross section is shown to the right. State whether the following could be the solid:



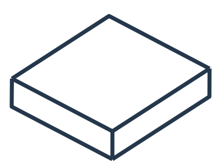
a



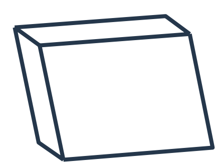
b



c



d



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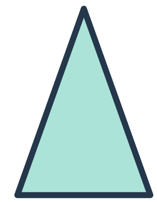
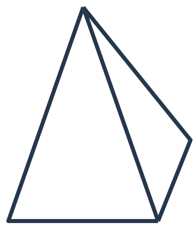
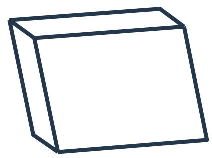
11 The shape shown is a cross section of a solid. State whether the following could be the solid:



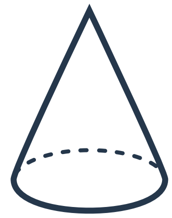
a



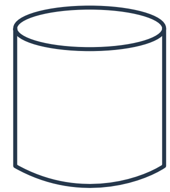
b



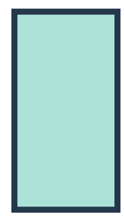
c



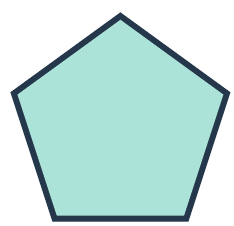
d



12 A solid is cut parallel to one of its faces to give the cross section shown to the right. What could the solid possibly be?



13 Specify what solids could have the following cross section:



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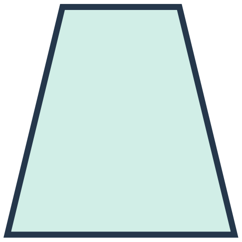
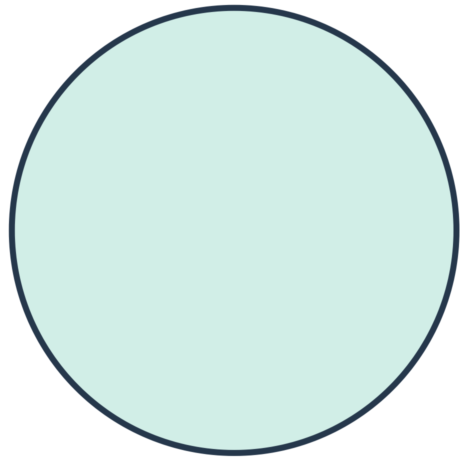
14 Describe and sketch a solid that could have the following shapes as its cross section:

a



b



Applications

15 Hiroki claims that it doesn't matter which way you take the cross section of a sphere, the cross section will always be same shape.
Is Hiroki correct? Can you think of any other solids which have this property?

16 Determine whether the following statement is true:
"The side view of a shape is always the same shape as the perpendicular cross section."
Give an example which justifies your response.

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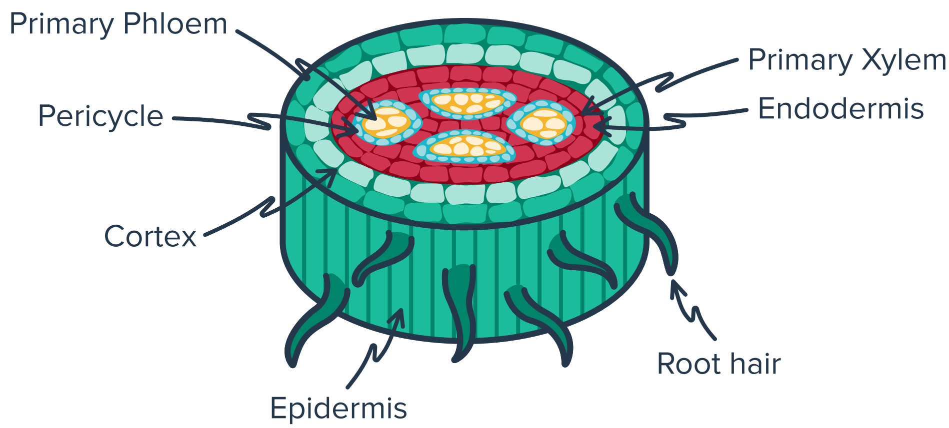
17 In Biology class, students performed a lab experiment where they looked at the perpendicular cross section of a root under a microscope:

Cross Section of a Root



If you wish to study all of the cellular components of a root, would you choose to take examine a cross section that is parallel or perpendicular to the epidermis? Explain your answer.

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